

VisionInspect AI: Manufacturing Defect Detection & Quality Inspection System

Infosys Springboard Internship 7.0

Milestone 1

Team 1

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1.Introduction

VisionInspectAI is an AI-based project that focuses on detecting defects in industrial products using image data.

Before building any machine learning model, it is important to understand the dataset and ensure it is properly prepared. This report documents the work I completed for Milestone 1, which includes dataset exploration, preprocessing, statistical analysis and visualization. These steps help verify the quality of the dataset and provide a strong foundation for the next stages of the project.

2. Dataset Description

2.1 Dataset Overview

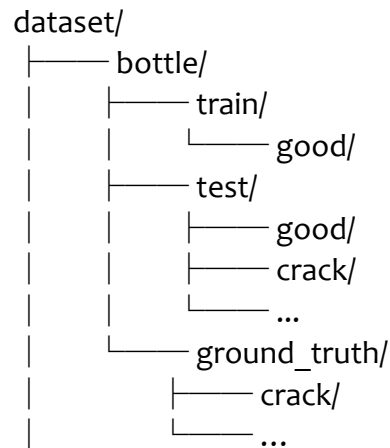
The project uses the MVTec Anomaly Detection (MVTec AD) dataset, a benchmark dataset for industrial anomaly detection. It contains images of 15 object and texture categories, each organized into train, test, and ground truth folders. The training set consists of defect-free images, while the test set includes both normal and defective samples. Ground truth masks are provided for defective images to identify the exact defect regions.

The dataset structure enables systematic exploration, preprocessing, statistical analysis, and visualization before model development. The categories explored include bottle, cable, capsule, carpet, grid, hazelnut, leather, metal nut, pill, screw, tile, toothbrush, transistor, wood, and zipper.

2.2 Dataset Structure

The dataset is organized into separate folders for each category. Every category contains train, test and ground_truth directories. The train folder contains normal images, the test folder contains both normal and defective images, and the

ground_truth folder stores defect masks corresponding to defective test images.



2.3 Dataset Statistics

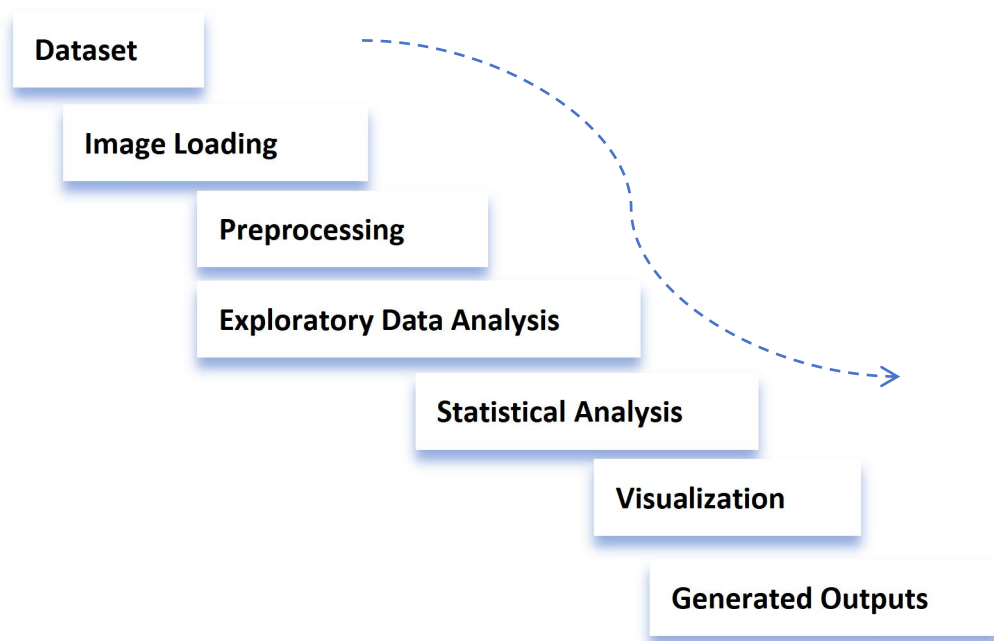
Category	Train Images	Test Images	Ground Truth Masks
Bottle	209	83	63
Cable	224	150	92
Capsule	219	132	109
Carpet	280	117	89
Grid	264	78	57
Hazelnut	391	110	70
Leather	245	124	92
Metal Nut	220	115	93
Pill	267	167	141
Screw	320	160	119
Tile	230	117	84
Toothbrush	60	42	30
Transistor	213	100	40
Wood	247	79	60
Zipper	240	151	119

Table: Category statistics output

- The dataset consists of 15 object and texture categories and folders. The number of images varies across categories, providing a diverse set of normal and defective samples for analysis and model development.

3. Methodology

3.1 WorkFlow



3.2 Image Loading

Images from each category were loaded from the dataset directory. The train, test and ground truth folders were accessed separately to verify the dataset organization and image availability before further processing.

3.3 Data Preprocessing

The images were re-sized to a fixed resolution and converted into a normalized format for consistent analysis. The preprocessing step also verified image dimensions and ensured that the dataset could be processed without inconsistencies.

3.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to examine the dataset structure and distribution. Image counts, category-wise organization, and sample images were analyzed to better understand the characteristics of the dataset.

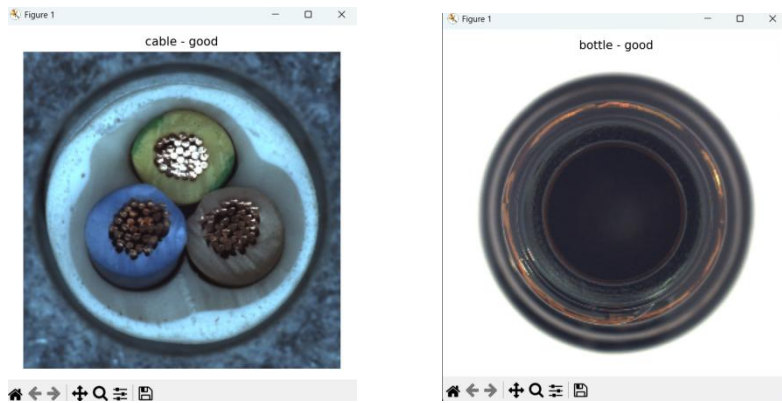


Fig:Sample Images of the Categories (Cable, Bottle)

3.5 Statistical Analysis

Statistical summaries were generated for each category, including image counts, defect distributions, and percentage-wise analysis. These statistics provide a quantitative understanding of the dataset before model development.

3.6 Data Visualization

Visualizations were created using Matplotlib to represent category distributions, defect statistics, and processed sample images. These charts provide a clear overview of the dataset and help validate the preprocessing results.

4 Implementation

4.1 preprocessing module

Purpose

Handles image preprocessing before analysis by preparing the dataset in a consistent format.

Implementation

- Images were loaded using openCV
- Images were re-sized to 256 x 256 pixels
- Pixel values were normalized into the range [0,1]
- Preprocessed images were verified by checking shape, datatype, and pixel value range before further analysis.

```
Original-Image Shape : (900, 900, 3)
```

```
Processed Image  
Shape : (256, 256, 3)  
Dtype : float32  
Min : 0.1137  
Max : 1.0000
```



Fig Preprocessed Image

4.2 EDA module

Purpose

Performs Exploratory Data Analysis on the dataset.

Implementation

- Reads each dataset category
- Examines folder structure
- Counts training, testing and ground truth images.
- Displays sample images
- Generates category-wise summaries.

4.3 statistics module

Purpose

Generates statistical summaries for all dataset categories.

Implementation

- Calculates image counts.
 - Computes defect-wise percentages.
 - Produces category-level statistics.
- ```
def get_category_statistics(category_path):
 train = count_images(category_path / "train")
 test = count_images(category_path / "test")
 ground_truth = count_images(category_path / "ground_truth")
```

## 4.4 Visualization module

### **Purpose**

Creates graphical representations of the dataset.

### **Implementation**

- Generates bar charts.
- Displays category distributions
- Visualizes defect statistics.
- Saves output charts for further analysis.

## 4.5 Execution Pipeline

### **Purpose**

Acts as the entry point for executing the preprocessing, analysis , statistics and visualization modules.

### **Implementation**

- Calls the required functions
- Executes the complete analysis pipeline
- Displays the generated outputs.

## 4 Results and Observations

The implemented preprocessing and EDA pipeline successfully analyzed all selected dataset categories. The generated statistics and visualizations provided a clear understanding of the dataset composition and defect distribution.

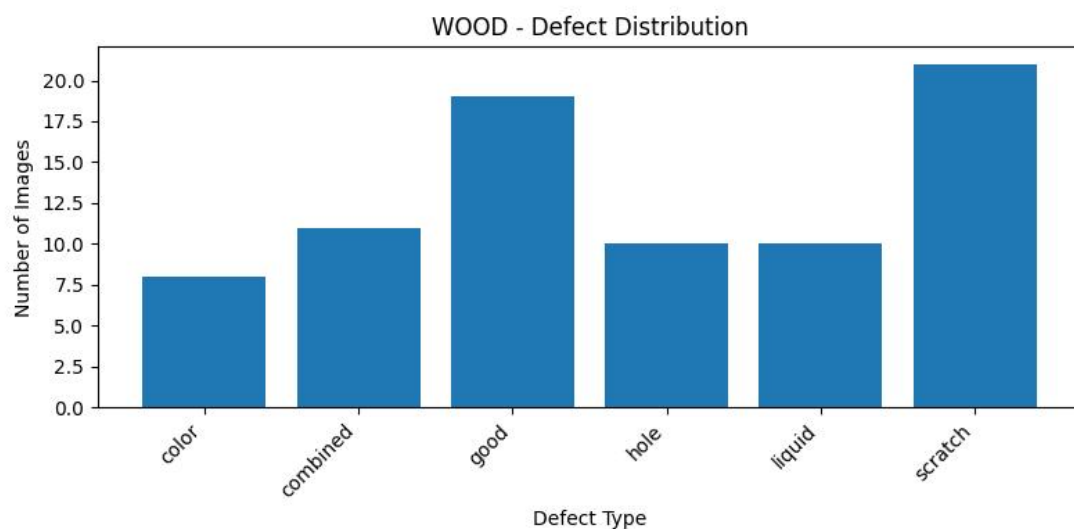


Fig: Defect Distribution Bar Chart (Category: Wood)

- The visualization indicates that defect counts vary significantly across categories, with some categories containing a higher proportion of defective samples than others.

### **Key Observations**

- ✓ The dataset is organized into 15 object categories, each containing train, test, and ground\_truth folders.
- ✓ Image preprocessing produced a consistent output size of  $256 \times 256 \times 3$ , making the dataset suitable for further model training.
- ✓ Category-wise image counts and defect distributions were successfully generated.
- ✓ Defect percentages varied across categories, highlighting differences in the number of defective and normal samples.
- ✓ Bar charts and statistical summaries made it easier to compare categories and understand the dataset before model development.
- ✓ The generated outputs can serve as a reference for future preprocessing, model training, and evaluation.

## **5 Conclusion**

The initial dataset analysis, preprocessing, and statistical exploration were successfully completed for the project. The implemented pipeline verifies the dataset structure, processes images into a consistent format, and generates category-wise statistics and visualizations. These outputs provide a clear understanding of the dataset and establish a solid foundation for the subsequent model development and defect detection stages.

## 6. References

1. MVTec AD Dataset  
<https://www.mvtec.com/company/research/datasets/mvtec-ad>
2. Python Documentation  
<https://docs.python.org/3/>
3. OpenCV Documentation  
<https://docs.opencv.org/>
4. NumPy Documentation  
<https://numpy.org/doc/>
5. Matplotlib Documentation  
<https://matplotlib.org/>
6. VisionInspectAI GitHub Repository  
<https://github.com/GKSJ-Deepvision/VisionInspectAI>